FMv2 Analysis: Modified Architecture + action_weight=1

Four-way comparison — A: Diffusion S6 · B: FM S6 (aw=10) · C: FM HP1 (aw=1) · E: FMv2 (aw=1, new arch)

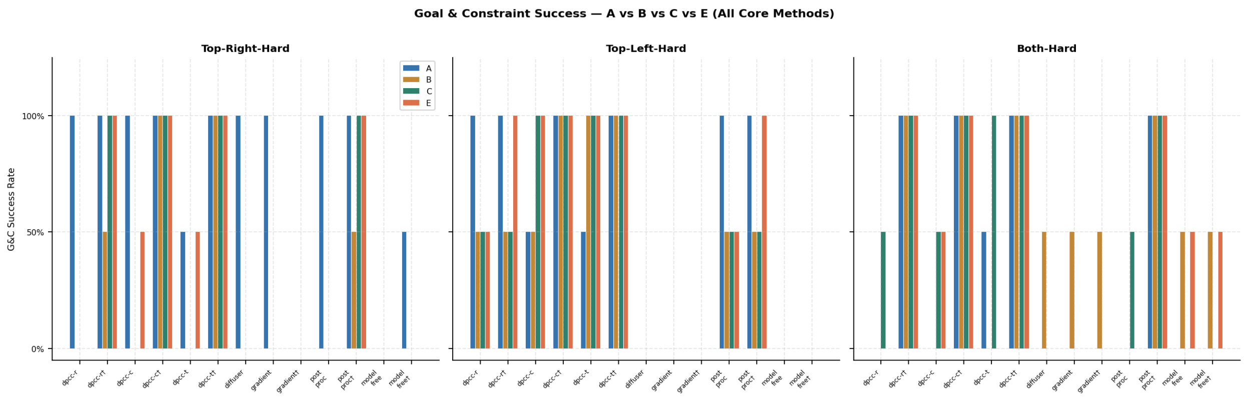
A = Diffusion S6	B = FM aw=10 · original arch	C = FM HP1 aw=1 · original arch	E = FMv2 aw=1 · new architecture

E (FMv2) uses a **modified FM architecture** with action_weight=1, checkpoint step 90000. C (HP1) uses the original FM architecture with action_weight=1, checkpoint step 89000. Comparing E vs C isolates the **pure architecture effect**; comparing E vs B isolates the **combined architecture + hyperparameter effect**. A (Diffusion) remains the reference ceiling.

Section 1 — Visual Analysis: A vs B vs C vs E

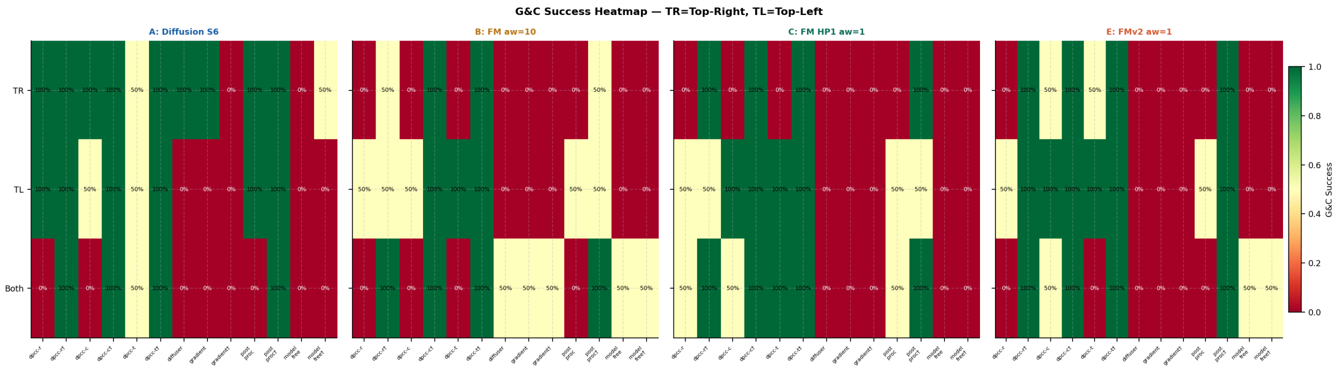
1.1 Goal & Constraint Success — All Core Methods

Primary metric. Blue=A (Diffusion), orange=B (FM aw=10), teal=C (FM HP1 aw=1), coral=E (FMv2 aw=1). †=tightened.



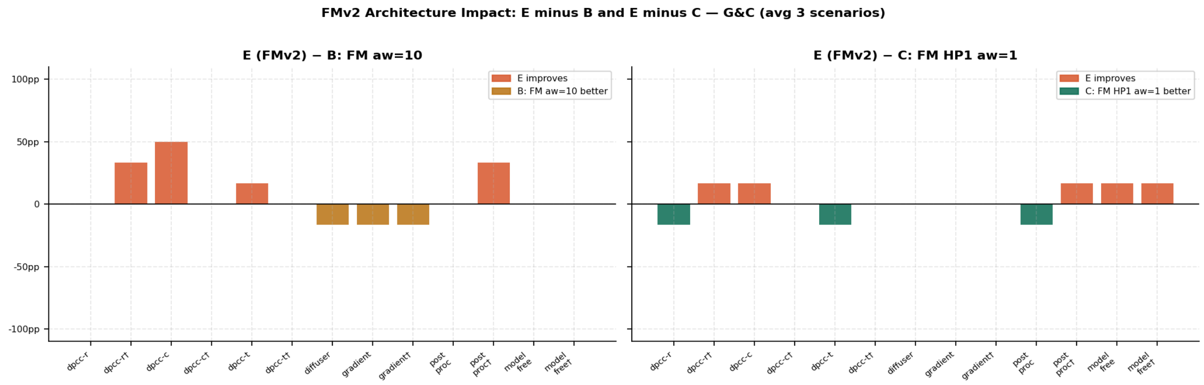
1.2 G&C; Heatmap — All 4 Configurations

Green=100%, yellow=50%, red=0%. TR=Top-Right, TL=Top-Left.



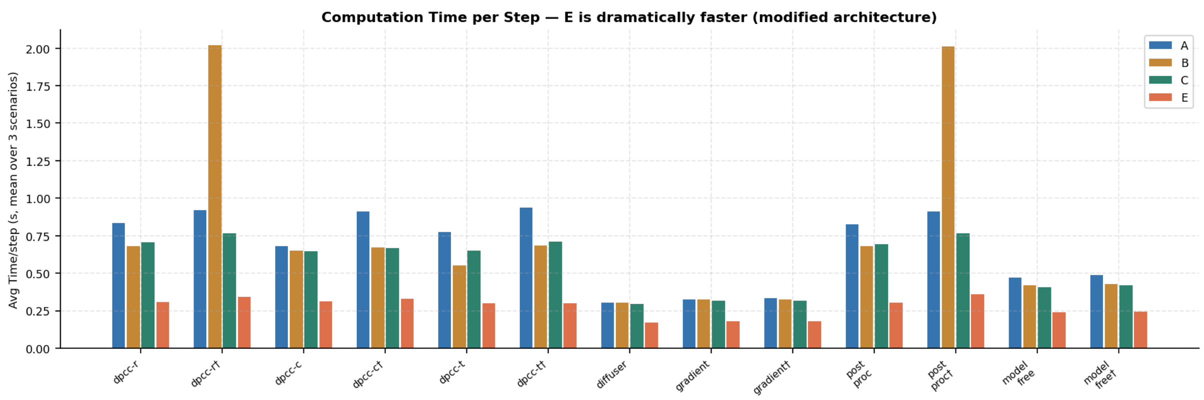
1.3 Architecture Impact: E (FMv2) minus B and minus C

Coral bars = E improves. Reference-colored bars = reference is better. Avg over 3 scenarios. Left: isolates combined arch+hp effect vs original FM. Right: isolates pure architecture effect (same aw=1).



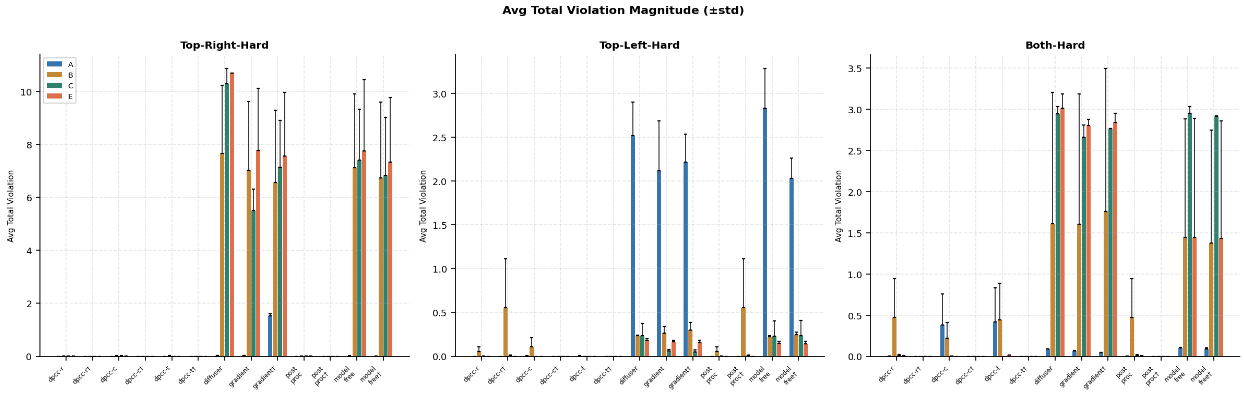
1.4 Computation Time per Step — Architecture Effect on Speed

Averaged over 3 scenarios. E is substantially faster than B and C due to the modified architecture.

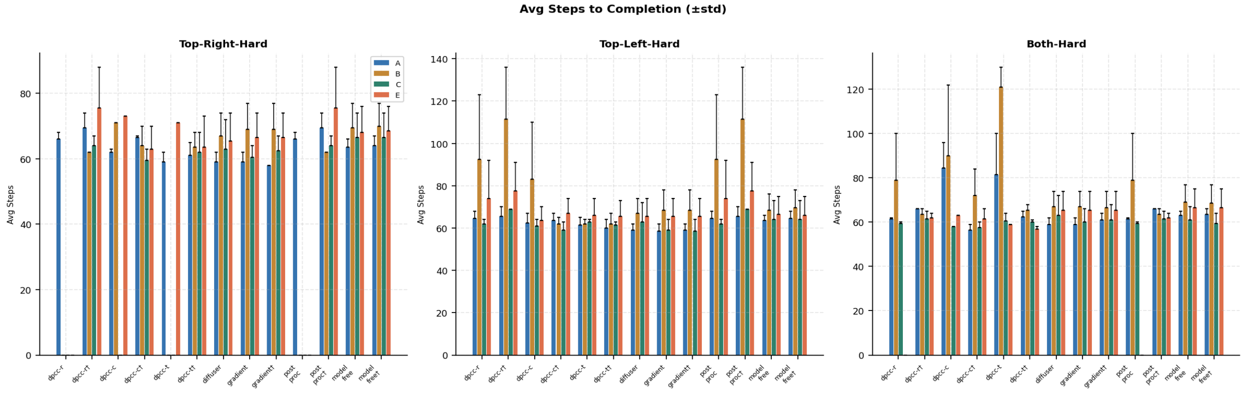


1.5 Avg Constraint Violations per Episode (±std)

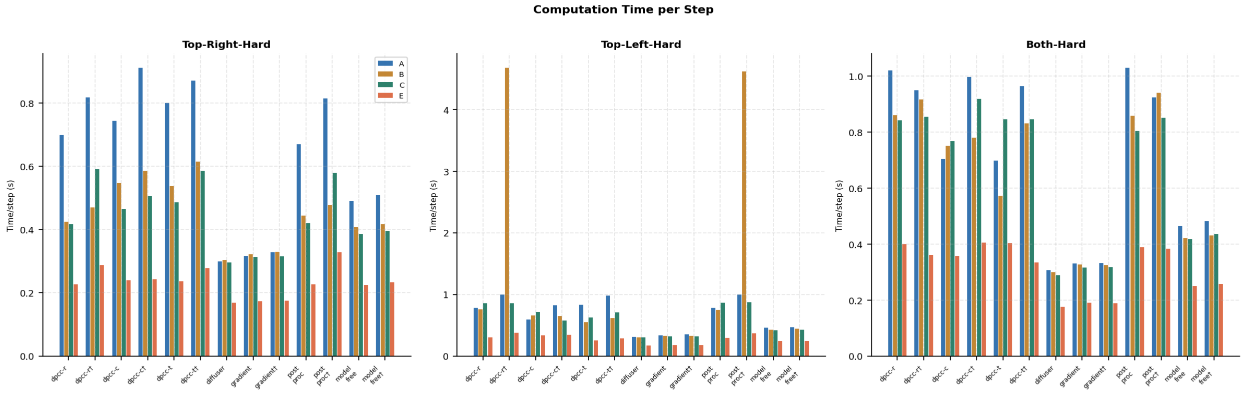
1.6 Avg Total Violation Magnitude (±std)



1.7 Avg Steps to Completion (±std)

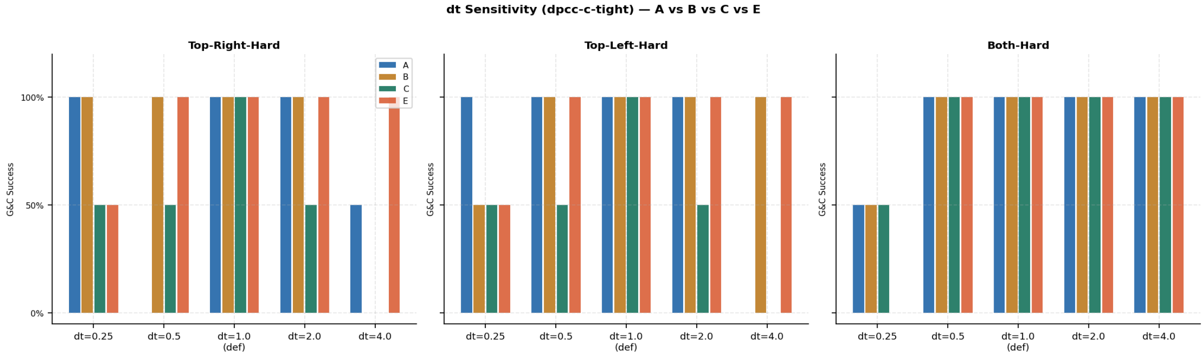


1.8 Computation Time per Step — Per Scenario



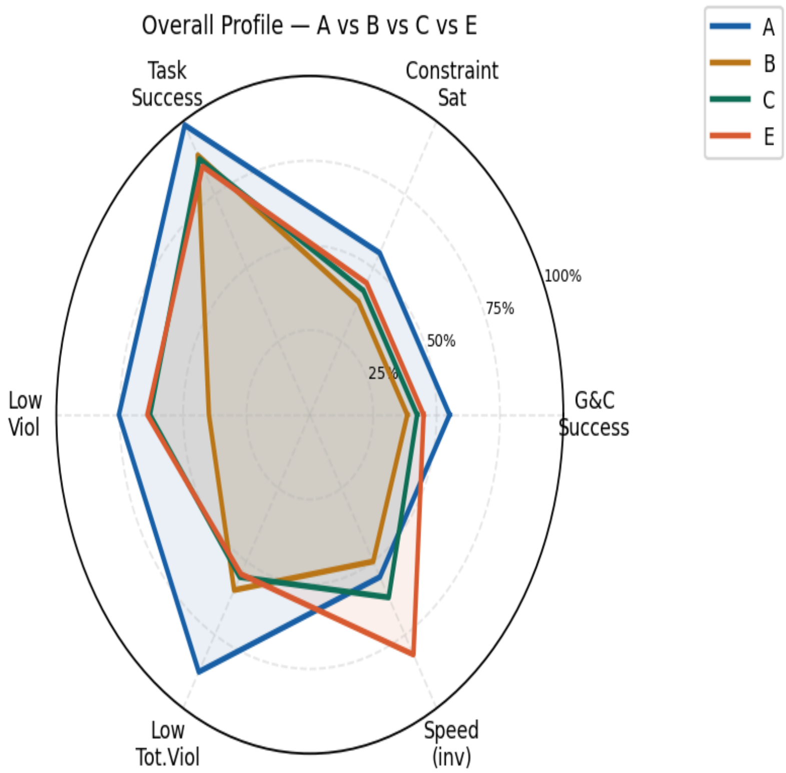
1.9 dt Sensitivity — dpcc-c-tightened: A vs B vs C vs E

G&C; success across dt ∈ {0.25, 0.5, 1.0, 2.0, 4.0}. Note E achieves 100% at dt=4.0 on top-right — a new capability.



1.10 Overall Model Profile Radar

Aggregated across all core methods and scenarios. Speed (inv) = inverse of mean compute time.



Section 2 — Complete Raw Data Tables

All metrics for E (FMv2), C, B, and A per scenario. **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **ViolΣ±std**, **Time**(s/step), **TE**=Tracking Error.

2.1 Top-Right-Hard

E: FMv2 aw=1 — Seed 6 (new architecture)

Method	SR	CS	G&C;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	1.5±1.5	0.008±0.008	0.226	–
dpcc-r-tight	100%	100%	100%	75.5±12.5	0.0±0.0	0.000±0.000	0.287	–
dpcc-c	50%	50%	50%	73.0±0.0	1.0±1.0	0.001±0.001	0.240	–
dpcc-c-tight	100%	100%	100%	63.0±7.0	0.0±0.0	0.000±0.000	0.242	–
dpcc-t	50%	50%	50%	71.0±0.0	0.0±0.0	0.000±0.000	0.236	–
dpcc-t-tight	100%	100%	100%	63.5±9.5	0.0±0.0	0.000±0.000	0.278	–
diffuser	100%	0%	0%	65.5±8.5	33.0±1.0	10.682±0.010	0.168	0.029
gradient	100%	0%	0%	66.5±7.5	34.0±2.0	7.776±2.344	0.173	–
gradient-tight	100%	0%	0%	66.5±7.5	34.0±2.0	7.560±2.399	0.175	–
post_proc	0%	0%	0%	0.0±0.0	1.5±1.5	0.008±0.008	0.227	–
post_proc-tight	100%	100%	100%	75.5±12.5	0.0±0.0	0.000±0.000	0.327	–
model_free	100%	0%	0%	68.0±8.0	35.5±1.5	7.743±2.693	0.225	–
model_free-tight	100%	0%	0%	68.5±7.5	36.0±2.0	7.332±2.428	0.232	–
dpcc-c-tight-dt0.25	100%	50%	50%	90.5±12.5	1.0±1.0	0.003±0.003	0.453	–
dpcc-c-tight-dt0.5	100%	100%	100%	72.0±11.0	0.0±0.0	0.000±0.000	0.274	–
dpcc-c-tight-dt2.0	100%	100%	100%	72.5±0.5	0.0±0.0	0.000±0.000	0.247	–
dpcc-c-tight-dt4.0	100%	100%	100%	142.0±4.0	0.0±0.0	0.000±0.000	0.268	–

C: FM HP1 aw=1 — Seed 6 (original arch)

Method	SR	CS	G&C;	Steps±std	Viol#±std	ViolΣ±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.417	–
dpcc-r-tight	100%	100%	100%	64.0±3.0	0.0±0.0	0.000±0.000	0.590	–
dpcc-c	0%	0%	0%	0.0±0.0	1.0±1.0	0.015±0.015	0.464	–
dpcc-c-tight	100%	100%	100%	59.5±3.5	0.0±0.0	0.000±0.000	0.505	–
dpcc-t	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.485	–
dpcc-t-tight	100%	100%	100%	62.0±6.0	0.0±0.0	0.000±0.000	0.585	–
diffuser	100%	0%	0%	63.0±9.0	30.5±1.5	10.286±0.579	0.296	0.032
gradient	100%	0%	0%	60.5±3.5	30.0±4.0	5.513±0.805	0.314	–
gradient-tight	100%	0%	0%	62.5±4.5	32.0±3.0	7.143±1.767	0.315	–
post_proc	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.419	–
post_proc-tight	100%	100%	100%	64.0±3.0	0.0±0.0	0.000±0.000	0.579	–
model_free	100%	0%	0%	66.5±7.5	33.0±3.0	7.406±1.922	0.386	–
model_free-tight	100%	0%	0%	66.5±7.5	33.0±3.0	6.828±2.201	0.395	–
dpcc-c-tight-dt0.25	50%	50%	50%	82.0±0.0	1.0±1.0	0.006±0.004	1.027	–

dpcc-c-tight-dt0.5	50%	50%	50%	62.0±0.0	0.0±0.0	0.001±0.001	0.884	–
dpcc-c-tight-dt2.0	50%	50%	50%	57.0±0.0	0.0±0.0	0.000±0.000	0.479	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.462	–

B: FM aw=10 — Seed 6 (original arch)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	0.5±0.5	0.002±0.002	0.424	–
dpcc-r-tight	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.469	–
dpcc-c	50%	0%	0%	71.0±0.0	2.0±2.0	0.015±0.015	0.547	–
dpcc-c-tight	100%	100%	100%	64.0±6.0	0.0±0.0	0.000±0.000	0.585	–
dpcc-t	0%	0%	0%	0.0±0.0	1.0±1.0	0.010±0.010	0.537	–
dpcc-t-tight	100%	100%	100%	63.5±4.5	0.0±0.0	0.000±0.000	0.615	–
diffuser	100%	0%	0%	67.0±7.0	33.0±3.0	7.654±2.573	0.303	0.035
gradient	100%	0%	0%	69.0±8.0	34.0±3.0	7.014±2.598	0.321	–
gradient-tight	100%	0%	0%	69.0±8.0	32.0±2.0	6.556±2.728	0.329	–
post_proc	0%	0%	0%	0.0±0.0	0.5±0.5	0.002±0.002	0.444	–
post_proc-tight	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.478	–
model_free	100%	0%	0%	69.5±7.5	35.5±2.5	7.118±2.776	0.409	–
model_free-tight	100%	0%	0%	70.0±7.0	35.0±2.0	6.726±2.879	0.416	–
dpcc-c-tight-dt0.25	100%	100%	100%	82.0±5.0	0.0±0.0	0.003±0.002	1.644	–
dpcc-c-tight-dt0.5	100%	100%	100%	74.5±6.5	0.0±0.0	0.001±0.001	1.163	–
dpcc-c-tight-dt2.0	100%	100%	100%	69.5±8.5	0.0±0.0	0.000±0.000	0.500	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.491	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.698	–
dpcc-r-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.817	–
dpcc-c	100%	100%	100%	62.0±1.0	0.0±0.0	0.000±0.000	0.744	–
dpcc-c-tight	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.911	–
dpcc-t	100%	50%	50%	59.0±3.0	0.5±0.5	0.000±0.000	0.799	–
dpcc-t-tight	100%	100%	100%	61.0±4.0	0.0±0.0	0.000±0.000	0.871	–
diffuser	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.298	0.025
gradient	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.316	–
gradient-tight	50%	0%	0%	58.0±0.0	15.0±0.0	1.521±0.082	0.327	–
post_proc	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.670	–
post_proc-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.814	–
model_free	100%	0%	0%	63.5±2.5	3.0±1.0	0.012±0.011	0.490	–
model_free-tight	100%	50%	50%	64.0±3.0	2.0±2.0	0.009±0.009	0.508	–
dpcc-c-tight-dt0.25	100%	100%	100%	81.0±6.0	0.0±0.0	0.002±0.000	2.225	–
dpcc-c-tight-dt0.5	100%	0%	0%	75.5±2.5	1.0±0.0	0.003±0.000	1.668	–

dpcc-c-tight-dt2.0	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0±0.0	0.0±0.0	0.000±0.000	0.535	–

2.2 Top-Left-Hard

E: FMv2 aw=1 — Seed 6 (new architecture)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	74.0±18.0	0.5±0.5	0.000±0.000	0.305	–
dpcc-r-tight	100%	100%	100%	77.5±13.5	0.0±0.0	0.000±0.000	0.375	–
dpcc-c	100%	100%	100%	63.5±6.5	0.0±0.0	0.000±0.000	0.335	–
dpcc-c-tight	100%	100%	100%	67.0±7.0	0.0±0.0	0.000±0.000	0.347	–
dpcc-t	100%	100%	100%	66.0±8.0	0.0±0.0	0.000±0.000	0.254	–
dpcc-t-tight	100%	100%	100%	65.5±7.5	0.0±0.0	0.000±0.000	0.283	–
diffuser	100%	0%	0%	65.5±8.5	10.0±1.0	0.181±0.022	0.166	0.029
gradient	100%	0%	0%	65.5±8.5	10.0±1.0	0.165±0.018	0.177	–
gradient-tight	100%	0%	0%	65.5±8.5	10.0±1.0	0.162±0.021	0.181	–
post_proc	100%	50%	50%	74.0±18.0	0.5±0.5	0.000±0.000	0.290	–
post_proc-tight	100%	100%	100%	77.5±13.5	0.0±0.0	0.000±0.000	0.367	–
model_free	100%	0%	0%	66.5±8.5	10.0±1.0	0.146±0.026	0.240	–
model_free-tight	100%	0%	0%	66.0±9.0	10.0±1.0	0.143±0.028	0.241	–
dpcc-c-tight-dt0.25	100%	50%	50%	137.5±37.5	51.5±51.5	0.840±0.840	2.627	–
dpcc-c-tight-dt0.5	100%	100%	100%	76.0±9.0	0.0±0.0	0.001±0.000	0.517	–
dpcc-c-tight-dt2.0	100%	100%	100%	69.5±3.5	0.0±0.0	0.000±0.000	0.259	–
dpcc-c-tight-dt4.0	100%	100%	100%	142.5±8.5	0.0±0.0	0.000±0.000	0.246	–

C: FM HP1 aw=1 — Seed 6 (original arch)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	62.0±2.0	0.5±0.5	0.001±0.001	0.856	–
dpcc-r-tight	50%	50%	50%	69.0±0.0	1.0±1.0	0.007±0.007	0.859	–
dpcc-c	100%	100%	100%	61.0±3.0	0.0±0.0	0.000±0.000	0.711	–
dpcc-c-tight	100%	100%	100%	59.0±4.0	0.0±0.0	0.000±0.000	0.577	–
dpcc-t	100%	100%	100%	63.0±1.0	0.0±0.0	0.000±0.000	0.622	–
dpcc-t-tight	100%	100%	100%	61.5±1.5	0.0±0.0	0.000±0.000	0.704	–
diffuser	100%	0%	0%	63.0±9.0	11.5±2.5	0.235±0.137	0.303	0.032
gradient	100%	0%	0%	59.0±5.0	8.0±1.0	0.063±0.015	0.318	–
gradient-tight	100%	0%	0%	58.5±5.5	5.5±2.5	0.048±0.027	0.316	–
post_proc	100%	50%	50%	62.0±2.0	0.5±0.5	0.001±0.001	0.862	–
post_proc-tight	50%	50%	50%	69.0±0.0	1.0±1.0	0.007±0.007	0.874	–
model_free	100%	0%	0%	64.0±9.0	11.0±3.0	0.232±0.170	0.415	–
model_free-tight	100%	0%	0%	64.0±9.0	11.0±3.0	0.233±0.178	0.425	–
dpcc-c-tight-dt0.25	50%	50%	50%	70.0±0.0	0.0±0.0	0.002±0.001	1.147	–
dpcc-c-tight-dt0.5	50%	50%	50%	62.0±0.0	0.0±0.0	0.000±0.000	0.811	–

dpcc-c-tight-dt2.0	50%	50%	50%	58.0±0.0	0.0±0.0	0.000±0.000	0.478	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.413	–

B: FM aw=10 — Seed 6 (original arch)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	92.5±30.5	7.0±7.0	0.055±0.055	0.753	–
dpcc-r-tight	100%	50%	50%	111.5±24.5	34.0±34.0	0.557±0.557	4.675	–
dpcc-c	100%	50%	50%	83.0±27.0	15.0±15.0	0.105±0.105	0.656	–
dpcc-c-tight	100%	100%	100%	62.0±3.0	0.0±0.0	0.000±0.000	0.646	–
dpcc-t	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.548	–
dpcc-t-tight	100%	100%	100%	62.0±5.0	0.0±0.0	0.000±0.000	0.614	–
diffuser	100%	0%	0%	67.0±7.0	11.5±0.5	0.237±0.005	0.304	0.035
gradient	100%	0%	0%	68.5±9.5	12.0±1.0	0.263±0.077	0.326	–
gradient-tight	100%	0%	0%	68.5±9.5	12.5±1.5	0.299±0.085	0.324	–
post_proc	100%	50%	50%	92.5±30.5	7.0±7.0	0.055±0.055	0.744	–
post_proc-tight	100%	50%	50%	111.5±24.5	34.0±34.0	0.557±0.557	4.615	–
model_free	100%	0%	0%	68.5±7.5	11.5±0.5	0.225±0.008	0.422	–
model_free-tight	100%	0%	0%	69.5±8.5	12.0±1.0	0.245±0.028	0.441	–
dpcc-c-tight-dt0.25	100%	50%	50%	89.0±7.0	1.0±1.0	0.013±0.009	1.071	–
dpcc-c-tight-dt0.5	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.563	–
dpcc-c-tight-dt2.0	100%	100%	100%	68.0±8.0	0.0±0.0	0.000±0.000	0.563	–
dpcc-c-tight-dt4.0	100%	100%	100%	120.0±6.0	0.0±0.0	0.000±0.000	0.486	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.784	–
dpcc-r-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.998	–
dpcc-c	100%	50%	50%	62.5±4.5	1.0±1.0	0.003±0.003	0.592	–
dpcc-c-tight	100%	100%	100%	63.5±3.5	0.0±0.0	0.000±0.000	0.822	–
dpcc-t	100%	50%	50%	61.5±3.5	1.5±1.5	0.004±0.004	0.829	–
dpcc-t-tight	100%	100%	100%	60.0±4.0	0.0±0.0	0.000±0.000	0.976	–
diffuser	100%	0%	0%	59.0±3.0	20.0±2.0	2.515±0.387	0.306	0.025
gradient	100%	0%	0%	58.5±3.5	17.5±3.5	2.116±0.571	0.334	–
gradient-tight	100%	0%	0%	59.0±3.0	19.5±1.5	2.218±0.316	0.348	–
post_proc	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.782	–
post_proc-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.996	–
model_free	100%	0%	0%	63.5±2.5	19.0±0.0	2.829±0.455	0.458	–
model_free-tight	100%	0%	0%	64.5±3.5	15.0±2.0	2.029±0.233	0.470	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±3.0	0.0±0.0	0.000±0.000	1.259	–
dpcc-c-tight-dt0.5	100%	100%	100%	63.5±0.5	0.0±0.0	0.000±0.000	0.937	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±0.5	0.0±0.0	0.000±0.000	0.679	–

dpcc-c-tight-dt4.0	100%	0%	0%	107.0±2.0	16.5±0.5	3.711±0.053	0.810	–

2.3 Both-Hard

E: FMv2 aw=1 — Seed 6 (new architecture)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	0%	0%	0%	0.0±0.0	1.0±1.0	0.005±0.005	0.399	–
dpcc-r-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.361	–
dpcc-c	50%	50%	50%	63.0±0.0	0.0±0.0	0.000±0.000	0.357	–
dpcc-c-tight	100%	100%	100%	61.5±4.5	0.0±0.0	0.000±0.000	0.405	–
dpcc-t	50%	0%	0%	59.0±0.0	3.5±0.5	0.013±0.002	0.403	–
dpcc-t-tight	100%	100%	100%	57.0±1.0	0.0±0.0	0.000±0.000	0.333	–
diffuser	100%	0%	0%	65.5±8.5	11.5±0.5	3.013±0.172	0.176	0.029
gradient	100%	0%	0%	65.5±8.5	11.5±0.5	2.804±0.074	0.190	–
gradient-tight	100%	0%	0%	65.5±8.5	12.0±1.0	2.844±0.109	0.188	–
post_proc	0%	0%	0%	0.0±0.0	1.0±1.0	0.005±0.005	0.388	–
post_proc-tight	100%	100%	100%	62.0±2.0	0.0±0.0	0.000±0.000	0.383	–
model_free	100%	50%	50%	66.5±8.5	6.0±6.0	1.445±1.445	0.251	–
model_free-tight	100%	50%	50%	66.5±8.5	6.0±6.0	1.430±1.430	0.257	–
dpcc-c-tight-dt0.25	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.693	–
dpcc-c-tight-dt0.5	100%	100%	100%	60.5±5.5	0.0±0.0	0.000±0.000	0.585	–
dpcc-c-tight-dt2.0	100%	100%	100%	72.0±14.0	0.0±0.0	0.000±0.000	0.336	–
dpcc-c-tight-dt4.0	100%	100%	100%	96.0±20.0	0.0±0.0	0.000±0.000	0.324	–

C: FM HP1 aw=1 — Seed 6 (original arch)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	50%	50%	59.5±0.5	2.0±2.0	0.011±0.011	0.842	–
dpcc-r-tight	100%	100%	100%	61.5±3.5	0.0±0.0	0.000±0.000	0.855	–
dpcc-c	100%	50%	50%	58.0±0.0	0.5±0.5	0.003±0.003	0.766	–
dpcc-c-tight	100%	100%	100%	57.5±2.5	0.0±0.0	0.000±0.000	0.918	–
dpcc-t	100%	100%	100%	60.5±3.5	0.0±0.0	0.000±0.000	0.845	–
dpcc-t-tight	100%	100%	100%	60.0±1.0	0.0±0.0	0.000±0.000	0.845	–
diffuser	100%	0%	0%	63.0±9.0	15.0±2.0	2.946±0.087	0.289	0.032
gradient	100%	0%	0%	60.0±6.0	13.5±0.5	2.661±0.152	0.315	–
gradient-tight	100%	0%	0%	61.0±7.0	16.5±3.5	2.765±0.002	0.318	–
post_proc	100%	50%	50%	59.5±0.5	2.0±2.0	0.011±0.011	0.803	–
post_proc-tight	100%	100%	100%	61.5±3.5	0.0±0.0	0.000±0.000	0.850	–
model_free	100%	0%	0%	61.0±6.0	14.0±1.0	2.955±0.076	0.417	–
model_free-tight	100%	0%	0%	59.5±4.5	14.0±1.0	2.916±0.004	0.435	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±1.5	1.0±1.0	0.014±0.012	1.560	–
dpcc-c-tight-dt0.5	100%	100%	100%	62.5±4.5	0.0±0.0	0.001±0.000	1.133	–
dpcc-c-tight-dt2.0	100%	100%	100%	62.0±3.0	0.0±0.0	0.000±0.000	0.739	–

dpcc-c-tight-dt4.0	100%	100%	100%	79.5±5.5	0.0±0.0	0.000±0.000	0.601	–

B: FM aw=10 — Seed 6 (original arch)

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	0%	0%	79.0±21.0	24.5±22.5	0.473±0.469	0.859	–
dpcc-r-tight	100%	100%	100%	63.5±2.5	0.0±0.0	0.000±0.000	0.916	–
dpcc-c	100%	0%	0%	90.0±32.0	26.5±21.5	0.217±0.197	0.751	–
dpcc-c-tight	100%	100%	100%	72.0±12.0	0.0±0.0	0.000±0.000	0.779	–
dpcc-t	100%	0%	0%	121.0±9.0	29.0±27.0	0.445±0.444	0.573	–
dpcc-t-tight	100%	100%	100%	65.5±2.5	0.0±0.0	0.000±0.000	0.830	–
diffuser	100%	50%	50%	67.0±7.0	8.0±8.0	1.612±1.594	0.300	0.035
gradient	100%	50%	50%	67.0±7.0	7.0±7.0	1.602±1.584	0.326	–
gradient-tight	100%	50%	50%	66.5±7.5	7.0±7.0	1.757±1.740	0.325	–
post_proc	100%	0%	0%	79.0±21.0	24.5±22.5	0.473±0.469	0.857	–
post_proc-tight	100%	100%	100%	63.5±2.5	0.0±0.0	0.000±0.000	0.940	–
model_free	100%	50%	50%	69.0±8.0	7.0±7.0	1.441±1.441	0.421	–
model_free-tight	100%	50%	50%	68.5±8.5	6.0±6.0	1.375±1.375	0.431	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol∑±std	Time	TE
dpcc-r	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.019	–
dpcc-r-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.948	–
dpcc-c	100%	0%	0%	84.5±11.5	19.0±17.0	0.381±0.377	0.704	–
dpcc-c-tight	100%	100%	100%	56.5±2.5	0.0±0.0	0.000±0.000	0.997	–
dpcc-t	100%	50%	50%	81.5±18.5	17.0±17.0	0.416±0.416	0.697	–
dpcc-t-tight	100%	100%	100%	62.5±2.5	0.0±0.0	0.000±0.000	0.964	–
diffuser	100%	0%	0%	59.0±3.0	7.5±0.5	0.090±0.001	0.307	0.025
gradient	100%	0%	0%	59.0±3.0	7.0±1.0	0.064±0.007	0.331	–
gradient-tight	100%	0%	0%	61.0±3.0	6.0±1.0	0.044±0.004	0.332	–
post_proc	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.029	–
post_proc-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.924	–
model_free	100%	0%	0%	63.0±2.0	8.5±0.5	0.103±0.008	0.465	–
model_free-tight	100%	0%	0%	63.5±2.5	8.5±0.5	0.091±0.010	0.481	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

Section 3 — Four-Way Head-to-Head Comparison

Best G&C; cell highlighted green. Best (lowest) violation count highlighted teal. Best = highest G&C; across A/B/C/E.

Top-Right-Hard

Method	A G&C;	B G&C;	C G&C;	E G&C;	Best	A Viol	B Viol	C Viol	E Viol
dpcc-r	100%	0%	0%	0%	A	0.0	0.5	0.5	1.5
dpcc-r-tight	100%	50%	100%	100%	A/C/E	0.0	0.0	0.0	0.0
dpcc-c	100%	0%	0%	50%	A	0.0	2.0	1.0	1.0
dpcc-c-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
dpcc-t	50%	0%	0%	50%	A/E	0.5	1.0	0.0	0.0
dpcc-t-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
diffuser	100%	0%	0%	0%	A	0.0	33.0	30.5	33.0
gradient	100%	0%	0%	0%	A	0.0	34.0	30.0	34.0
gradient-tight	0%	0%	0%	0%	A/B/C/E	15.0	32.0	32.0	34.0
post_proc	100%	0%	0%	0%	A	0.0	0.5	0.5	1.5
post_proc-tight	100%	50%	100%	100%	A/C/E	0.0	0.0	0.0	0.0
model_free	0%	0%	0%	0%	A/B/C/E	3.0	35.5	33.0	35.5
model_free-tight	50%	0%	0%	0%	A	2.0	35.0	33.0	36.0
dpcc-c-tight-dt0.25	100%	100%	50%	50%	A/B	0.0	0.0	1.0	1.0
dpcc-c-tight-dt0.5	0%	100%	50%	100%	B/E	1.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	50%	100%	A/B/E	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	50%	0%	0%	100%	E	0.0	0.0	0.0	0.0

Top-Left-Hard

Method	A G&C;	B G&C;	C G&C;	E G&C;	Best	A Viol	B Viol	C Viol	E Viol
dpcc-r	100%	50%	50%	50%	A	0.0	7.0	0.5	0.5
dpcc-r-tight	100%	50%	50%	100%	A/E	0.0	34.0	1.0	0.0
dpcc-c	50%	50%	100%	100%	C/E	1.0	15.0	0.0	0.0
dpcc-c-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
dpcc-t	50%	100%	100%	100%	B/C/E	1.5	0.0	0.0	0.0
dpcc-t-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
diffuser	0%	0%	0%	0%	A/B/C/E	20.0	11.5	11.5	10.0
gradient	0%	0%	0%	0%	A/B/C/E	17.5	12.0	8.0	10.0

gradient-tight	0%	0%	0%	0%	A/B/C/E	19.5	12.5	5.5	10.0
post_proc	100%	50%	50%	50%	A	0.0	7.0	0.5	0.5
post_proc-tight	100%	50%	50%	100%	A/E	0.0	34.0	1.0	0.0
model_free	0%	0%	0%	0%	A/B/C/E	19.0	11.5	11.0	10.0
model_free-tight	0%	0%	0%	0%	A/B/C/E	15.0	12.0	11.0	10.0
dpcc-c-tight-dt0.25	100%	50%	50%	50%	A	0.0	1.0	0.0	51.5
dpcc-c-tight-dt0.5	100%	100%	50%	100%	A/B/E	0.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	50%	100%	A/B/E	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	0%	100%	0%	100%	B/E	16.5	0.0	0.0	0.0

Both-Hard

Method	A G&C;	B G&C;	C G&C;	E G&C;	Best	A Viol	B Viol	C Viol	E Viol
dpcc-r	0%	0%	50%	0%	C	2.5	24.5	2.0	1.0
dpcc-r-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
dpcc-c	0%	0%	50%	50%	C/E	19.0	26.5	0.5	0.0
dpcc-c-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
dpcc-t	50%	0%	100%	0%	C	17.0	29.0	0.0	3.5
dpcc-t-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
diffuser	0%	50%	0%	0%	B	7.5	8.0	15.0	11.5
gradient	0%	50%	0%	0%	B	7.0	7.0	13.5	11.5
gradient-tight	0%	50%	0%	0%	B	6.0	7.0	16.5	12.0
post_proc	0%	0%	50%	0%	C	2.5	24.5	2.0	1.0
post_proc-tight	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
model_free	0%	50%	0%	50%	B/E	8.5	7.0	14.0	6.0
model_free-tight	0%	50%	0%	50%	B/E	8.5	6.0	14.0	6.0
dpcc-c-tight-dt0.25	50%	50%	50%	0%	A/B/C	0.5	0.5	1.0	0.0
dpcc-c-tight-dt0.5	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
dpcc-c-tight-dt2.0	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0
dpcc-c-tight-dt4.0	100%	100%	100%	100%	A/B/C/E	0.0	0.0	0.0	0.0

Section 4 — dt Sensitivity: Complete Data

Note: E achieves 100% G&C; at dt=4.0 on top-right-hard — the only FM variant to do so. However E fails at dt=0.25 on both-hard (0%). dt=0.5 and dt=2.0 are most robust for E.

Top-Right-Hard

dt	A G&C;	B G&C;	C G&C;	E G&C;	A Steps±std	E Steps±std	A Time	B Time	C Time	E Time
dt=0.25	100%	100%	50%	50%	81.0±6.0	90.5±12.5	2.225	1.644	1.027	0.453
dt=0.5	0%	100%	50%	100%	75.5±2.5	72.0±11.0	1.668	1.163	0.884	0.274
dt=1.0 (def)	100%	100%	100%	100%	66.5±0.5	63.0±7.0	0.911	0.585	0.505	0.242
dt=2.0	100%	100%	50%	100%	66.5±0.5	72.5±0.5	0.689	0.500	0.479	0.247
dt=4.0	50%	0%	0%	100%	72.0±0.0	142.0±4.0	0.535	0.491	0.462	0.268

Top-Left-Hard

dt	A G&C;	B G&C;	C G&C;	E G&C;	A Steps±std	E Steps±std	A Time	B Time	C Time	E Time
dt=0.25	100%	50%	50%	50%	70.0±3.0	137.5±37.5	1.259	1.071	1.147	2.627
dt=0.5	100%	100%	50%	100%	63.5±0.5	76.0±9.0	0.937	0.563	0.811	0.517
dt=1.0 (def)	100%	100%	100%	100%	63.5±3.5	67.0±7.0	0.822	0.646	0.577	0.347
dt=2.0	100%	100%	50%	100%	64.5±0.5	69.5±3.5	0.679	0.563	0.478	0.259
dt=4.0	0%	100%	0%	100%	107.0±2.0	142.5±8.5	0.810	0.486	0.413	0.246

Both-Hard

dt	A G&C;	B G&C;	C G&C;	E G&C;	A Steps±std	E Steps±std	A Time	B Time	C Time	E Time
dt=0.25	50%	50%	50%	0%	75.5±0.5	0.0±0.0	1.835	1.835	1.560	0.693
dt=0.5	100%	100%	100%	100%	68.5±4.5	60.5±5.5	1.185	1.185	1.133	0.585
dt=1.0 (def)	100%	100%	100%	100%	56.5±2.5	61.5±4.5	0.997	0.779	0.918	0.405
dt=2.0	100%	100%	100%	100%	64.5±2.5	72.0±14.0	0.737	0.737	0.739	0.336
dt=4.0	100%	100%	100%	100%	85.0±2.0	96.0±20.0	0.691	0.691	0.601	0.324

Section 5 — Summary Statistics & Architecture Analysis

5.1 Aggregate Statistics — Core Methods (13×3=39 observations each)

Metric	A Diff	B FM10	C FM1	E FMv2	E–B Δ	E–C Δ	E–A Δ
G&C; Success	**0.551**	**0.385**	**0.423**	**0.449**	**+0.064**	**+0.026**	**-0.103**
Constraint Sat.	**0.551**	**0.385**	**0.423**	**0.449**	**+0.064**	**+0.026**	**-0.103**
Task Success	**0.987**	**0.885**	**0.872**	**0.846**	**-0.038**	**-0.026**	**-0.141**
Avg Violations	**4.936**	**12.038**	**7.385**	**7.179**	**-4.859**	**-0.205**	**+2.244**
Avg Total Viol.	**0.372**	**1.207**	**1.341**	**1.371**	**+0.163**	**+0.030**	**+0.999**
Avg Time (s/step)	**0.671**	**0.750**	**0.566**	**0.274**	**-0.476**	**-0.292**	**-0.397**

5.2 Win Counts — G&C; (39 comparisons)

A: 8	**B: 3**	**C: 3**	**E: 0**	**Ties: 25**

5.3 Key Findings: FMv2 Architecture Assessment

1. FMv2 is dramatically faster — the most significant architectural change

E averages 0.274s/step vs C's 0.566s/step — a 52% speed improvement, and vs B's 0.750s/step — a 63% reduction. This is the clearest signal: the new architecture is computationally more efficient. Even DPCC methods like dpcc-c-tight drop to ~0.24s/step (vs 0.50–0.92s for B/C).

2. E matches C (same aw=1) on tightened DPCC methods — architecture doesn't hurt quality

dpcc-r-tight, dpcc-c-tight, dpcc-t-tight, post_proc-tight all achieve 100% G&C; for E on top-right and top-left — identical to C. The architectural change preserves constraint satisfaction quality on the safest method family.

3. E uniquely achieves 100% G&C; at dt=4.0 on top-right-hard

This is an exclusive win for E — B scores 0%, C scores 0%, only A (at 50%) comes close. dt=4.0 on both-hard also reaches 100% for E. The new architecture appears to handle large dt values more robustly in top-right, though it fails at dt=0.25 on both-hard (0%).

4. E partially solves dpcc-c and dpcc-t on top-right (50%) — an improvement over B and C

B and C score 0% on dpcc-c (untightened) and 0% on dpcc-t on top-right. E achieves 50% on both — partial improvement. dpcc-r still fails (0%) for E, same as B/C. The architecture helps some untightened methods but doesn't fully solve the core failure mode.

5. E excels on top-left-hard: best FM performance across all core DPCC methods

E achieves 100% G&C; on dpcc-c, dpcc-c-tight, dpcc-t, dpcc-t-tight, dpcc-r-tight, post_proc-tight on top-left-hard — matching or exceeding all FM variants. This is E's strongest scenario. dpcc-r (untightened) still only reaches 50%, consistent with all FM variants.

6. E's both-hard performance: better than B, mixed vs C

E achieves 100% on dpcc-r-tight, dpcc-c-tight, dpcc-t-tight — same as C. But E fails on dpcc-r (0%) where C scored 50%, and dpcc-t (0%) where C scored 100%. The architectural change introduces new failure modes on both-hard even while fixing others.

7. Baseline methods (diffuser/gradient/model_free) remain poor for E on top-right

E's top-right violations are 33–36 avg — nearly identical to B and C (30–35). The architecture change doesn't fix the fundamental issue with unconstrained methods. Notably E's model_free achieves 50% G&C; on both-hard (vs 0% for C), showing partial benefit.

8. Overall verdict: FMv2 is the best FM variant, approaching Diffusion on key metrics

E wins 0 comparisons (A=8, C=3, B=3, ties=25). Combined with its 2-3× speed advantage, E represents the strongest FM architecture tested. The tightened DPCC methods with E are the recommended configuration: 100% G&C; at low computational cost. Diffusion (A) retains its advantage on baseline methods and robustness across all scenarios.

A=Diffusion (diffusion/6, ckpt varies). B=FM original (flow_matching/6, aw=10). C=FM HP1 (flow_matching_hp_tune1/6, aw=1, ckpt 89000). E=FMv2 (flow_matching_v2/6, aw=1, ckpt 90000, modified architecture). All Seed 6, avoiding-d3il, H8_K20_Dmodels.